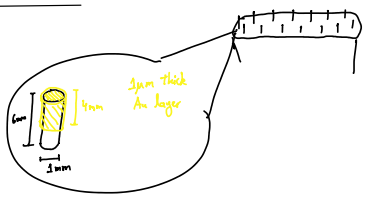
Hw 1

Wednesday, 18 January 2023 6:24 PM

Problem 1.3



Q1] Cost of gold in connector?

Volume of Au in 1 pin: $\pi (r_{pin} + Au_{thickness})^2 (l + 1\mu m) - \pi (r_{pin})^2 \cdot l$

$= \pi [(0.5\,mm + 1\mu m)^2] (3\,mm + 1\mu m) - \pi [0.5\,mm^2] \cdot 4\,mm$

$= \pi [(0.5\times10^{-3} + 1\times10^{-6})^2 (3.001\times10^{-3})] - \pi (0.5\times10^{-3})^2 \cdot 4\times10^{-3}$

$= 1.258\times10^{-11}\ m^3$

Total Volume: $1.258\times10^{-11}\ m^3$

$1\ cm = 10^{-2}\ m$

Mass of Au = Density of Au · Volume of Au

@LBMA UK [lbma.org.uk]: $\rho_{Au} = 19.32\ g/cm^3 = 19.32\times10^6\ g/m^3$

Mass of Au = $(19.32)(1.258\times10^{-5})\ g$

@MONEX Metals [monex.com/gold-prices/]: $\$61.38\ g^{-1}$

Total cost of gold for 1 pin = $(61.38)(19.32)(1.258\times10^{-5}) = \$0.015$ ⇒ (×25) Cost of Connector ⇒ $0.38

Q2] Cost/l of Au electrolyte (conc. = 40 mM)?

⇒ Conc. = $\frac{mol}{l}$; 1 l of electrolyte has 40×10^{-3} mol of Au

$MW_{Au} = 196.967\ \frac{g}{mol}$

⇒ Mass of Au (per l electrolyte): $40\times10^{-3}\cdot(196.967) = 7.878\ g/l$

⇒ Cost of electrolyte: $\$61.38\ \frac{}{g} \cdot \frac{7.878\ g}{l} = \$483.6$

Q3] Supporting electrolyte?

Adding another electrolyte will result in reduction in efficiency of the E-cell since another side reaction at counter electrode is likely to take place. This may allow for greater control of gold electroplating process?

Problem 1.4

Batteries vs Fuel Cells:

Similarities:

- Both batteries and fuel cells are devices that generate electricity via electrochemical reactions (i.e. transfer of electrons) at electrode interface. Both technologies depend on a species being oxidised at the anode resulting in flow of electrons around an external circuit that can then be used as voltage difference across a load.

Differences:

- A battery needs to be replenished by charging or adding in "electricity" to be stored for later use. A fuel cell needs chemical compounds (such as hydrogen in case of a hydrogen fuel cell) in order to provide electricity.
- A fuel cell operates as an open (continuous) system where it can produce electricity via the flowing fuel (hydrogen) and flow out of by product (such as water); a battery on the other hand, can provide DC current continuously until it runs (i.e. all lithium ions have been intercalated back into the cathode and need to be transported back to the graphitic anode to convert the battery into a charged state). Therefore a battery can be considered as a batch process, where it would need to ideally be stopped to charge again.

Links:
energy.gov
usdoe
Visual capitalist

Problem 1.7

Total Al production: 10^{11} lb/year
$= 4.54 \times 10^{10}$ kg/year

Faraday's law: relates charge passed to amount of product (Al) formed

$$M_i = \frac{-s_i (MW_i) Q}{nF}$$

; Q = total charge, s_i = stoich coeff, M_i = total mass formed, F = Faraday's constant

Al half reaction: $Al^{3+} + 3e^- \rightarrow Al$ (Cathode)

In 1 year:

$$4.54 \times 10^{13}\ g = \frac{(1)(26.98\ g/mol)\ Q}{(1)(96485\ C/mol)}$$

$$Q = 1.623 \times 10^{17}\ C$$

Total current: $I = \frac{Q}{t} = \frac{1.623 \times 10^{17}\ C}{1 \cdot 12 \cdot 30 \cdot 24 \cdot 60 \cdot 60} = 5.218 \times 10^{9}\ A$

$\therefore P = I \cdot V = 5.218 \times 10^{9}\ A \cdot 5\ V = 2.61 \times 10^{10}\ W$
$\approx 26\ GW$

Power consumption in an year $\approx$ 26 GW.

1 Watt-hour = 1 Watt · 1 hour ; 1 year = 8760 hours

↳ Global electric-power production (in 2000): IEA report estimates 12,696 TWh $\approx$ 1.449×10^{12} W in 1 year [@ iea.org]

Therefore, Al production consumed around 1.79% of total world power production

↳ Capacity of a larger power plant: [Ourworldindata.org]: 38000 MWh/day (typical coal plant in USA) Circa 2021

$= 1387000$ MWh/year

$= 158.33 \times 10^6$ W in 1 year

This value is 0.61% of what is needed in Al production

Problem 1.8

Half-cell reactions in electrochemical cell.

(a) $Cu(s) \mid CuSO_4(aq) \mid\mid CuSO_4(aq) \mid Cu(s)$

Left: $Cu^{2+} + 2e^- \rightleftharpoons Cu$

Right: $Cu \rightleftharpoons Cu^{2+} + 2e^-$

} No potential drop since same material is electrode

Overall: $Cu \rightleftharpoons Cu$

(b) $Ag(s) \mid AgCl(s) \mid HCl(aq) \mid H_2(g) \mid Pt(s)$

Left: $2Ag + 2Cl^- \rightleftharpoons 2AgCl_{(s)} + 2e^-$

Right: $H_2O + 2e^- + 2H^+ \rightleftharpoons H_{2(g)} + H_2O$

Overall: $2Ag + 2HCl \rightleftharpoons H_2 + 2AgCl$

(c) $Zn(s) \mid ZnO(s) \mid KOH(aq) \mid O_2(g) \mid Pt(s)$

Left: $2Zn_{(s)} + 4OH^- \rightleftharpoons 2ZnO + 4e^- + 2H_2O$

Right: $O_{2(g)} + 4e^- + 2H_2O \rightleftharpoons 4OH^-$ } Technically this should be a reduction?

Overall: $2Zn_{(s)} + O_{2(g)} \rightleftharpoons 2ZnO_{(s)}$

(d) $Pt(s) \mid O_2(g) \mid NaOH(aq) \mid NH_3(aq), NaNO_2(aq) \mid Pt(s)$

Left: $4OH^- \rightleftharpoons O_{2(g)} + 2H_2O + 4e^-$ $(\times \frac{3}{2})$

Right: $NO_2^- + 6e^- + 5H_2O \rightleftharpoons NH_3 + 7OH^-$

NH_4^+: $-3 \uparrow +4$; NO_2^-: $+3 \downarrow -4$

$[N^{3-} + 6e^- \rightleftharpoons N^{3+}]$

Overall: $NaNO_2 + 2H_2O \rightleftharpoons NH_3 + NaOH + \frac{3}{2}O_{2(g)}$

(e) $Pt(s) \mid FeCl_2(aq), FeCl_3(aq) \mid\mid SnCl_2(aq), SnCl_4(aq) \mid Pt(s)$

Left: $2FeCl_2 + 2Cl^- \rightleftharpoons 2FeCl_3 + 2e^-$ $[2Fe^{2+} \rightleftharpoons 2Fe^{3+} + 2e^-]$

Right: $SnCl_4 + 2e^- \rightleftharpoons SnCl_2 + 2Cl^-$ $[Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}]$

Overall: $2FeCl_2 + SnCl_4 \rightleftharpoons 2FeCl_3 + SnCl_2$

Reduction: $Cu^{2+} + 2e^- \rightarrow Cu$

Oxidation: $Cu \rightarrow Cu^{2+} + 2e^-$

Reduction: $H_2O + 2e^- \rightarrow H_2 + 2OH^-$

Oxidation: $Ag \rightarrow Ag^{2+} + 2e^-$

$[Ag^{2+} + Cl^- \rightarrow AgCl_{(s)}]$

Reduction: $H_2O + 2e^- \rightarrow O_2 +$

Oxidation: $Zn \rightarrow Zn^{2+} + 2e^-$

$[Zn^{2+}$

Problem 5

(a) Electron-conducting : Electrode 1
Electrode 2
Metal wire

Ion-conducting : Liquid Electrolyte

Insulating : Glass beaker

(b) Since e^- are flowing from V1 to V2 across a load R1, it is a galvanic cell

Oxidation @ V1 (loss of e^-) → Anode

Reduction @ V2 (gain e^-) → Cathode

Current flow in the opposite direction of e^- flow ($V2 \rightarrow V1$)

(c) When $V1 \approx V2$, there is no resistance to flow, therefore, we have max current (short circuit / closed circuit)

" $V1 \gg V2$, there is min current (negligible), and we have max cell voltage (open circuit potential) → V_{OC}

Max cell power is achieved at an intermediate value, where $\boxed{0 \leq V1 \leq V_{OC}}$

Problem 6

Bockris 1.1 – 1.5, 1.12

a. What did you know about electrochemistry before reading the chapter?
I knew that electrochemistry is essential to understanding how electrochemical cells (such as batteries) function. There are two types of electrochemical cells: galvanic and electrolytic.

b. What were 3 things you learned by reading the chapter? Even if ideas were new, were the physical and chemical concepts familiar to you?

- Firstly, it was illuminating to learn about the history of electrochemistry and its profound impact on technology years down the line, starting from Volta, Debye, Huckel, Faraday and Tafel.

- Field of electrochemistry can be subdivided into:
  - Ionics: ions in solution, solid solutions. This sub-field deals with:
    - Ion-solvent interaction,
    - Ion-ion interaction,
    - Ion transport interaction
    - Ionic liquids interaction
  - Electrodics: region between electronic and ionic conductor;
    - Interfacial region: electrically controlled surface reactions

- Electrochemical reaction, in its bare form, is just a transfer of electrons across an interface that results in a chemical tr ansformation. However, this chemical transformation occurs in a very different way, fundamentally, than a conventional chemical reaction. Reactants no longer coll ide with each other but with separated "charge transfer catalysts" (i.e. electrodes). From an overall perspective, the net reaction may have been the same but fundamentally electrochemical reactions differ from conventional chemical reactions. Furthermore, this spatial separation (among other things) gives electr ochemical cells an edge over other reactors since separation may not be an added cost in the production of chemicals.

- Reaction rate is the rate at which the power source pushes out and receives back electrons after their journey around the cir cuit. Therefore, one can easily control the electrochemical reaction rate: increasing the potential across the power source!

Yes the concepts were familiar but it was helpful to read them in an interesting thought experiment form to understand the fu ndamentals.

c. What were 2 things you are still confused about?
It is described that electrons are "physically" pushed from the power source into the electrode and then "overflow" into the solution to strike and neutralize ions in the layer of solution in contact with the metal. This may be a little misleading since it is sort of describing the electrons as "flowing" and then building up to then overflow to the solution.

My understanding is that the potential difference (due to the power source) results in increasing (or decreasing) the highest occupied molecular orbital in the electrode and therefore electrons want to transfer to the highest unoccupied molecular orbital of ions in solution that is at a lower (or higher) energy. This results in a transfer of electrons. Higher the potential difference (from the power source), the greater the difference between this molecular orbitals and faster the electron wants to jump across to lower energy.

d. Which of the actual or potential (section 1.12) applications of electrochemistry are most intriguing to you

It was interesting to learn the electrical dependence of humans increasing as time goes on where entire towns and cities would be self-sufficient in producing all that is required from electrochemical devices including even "internal fuel-cell powered heart" and electrically functioning body parts that will lead to a "person-machine combination" i.e. cyborgs. This seems pretty cool!